

```
In [1]: import pandas as pd
```

```
In [2]: dataset =pd.read_csv("customer_data.csv")
```

```
In [3]: dataset
```

```
Out[3]:
```

	name	age	gender	education	income	country	purchase_frequency
0	Teresa Williams MD	42	Female	High School	53936	Slovenia	0.9
1	Christine Myers	49	Female	Master	82468	Aruba	0.6
2	Dwayne Moreno	55	Male	Bachelor	56941	Cyprus	0.3
3	Amy Norton	24	Female	Bachelor	60651	Palau	0.2
4	Tonya Adams	64	Male	Master	81884	Zambia	0.9
...	...	...	...	...	...	...	...
995	Jason Haas	42	Female	PhD	98170	Bosnia and Herzegovina	0.7
996	Sharon Valdez	39	Male	High School	59174	Eritrea	0.9
997	Andrew Taylor	50	Male	Master	59808	Serbia	0.4
998	Jessica Johnson	50	Female	Master	58272	Cyprus	0.7
999	Brian Williams	38	Female	Bachelor	32512	Turkey	0.5

1000 rows × 8 columns

```
In [4]: dataset = dataset.drop(columns=['name', 'country'])
```

```
In [5]: dataset
```

```
Out[5]:
```

	age	gender	education	income	purchase_frequency	spending
0	42	Female	High School	53936	0.9	13227.120
1	49	Female	Master	82468	0.6	12674.040
2	55	Male	Bachelor	56941	0.3	5354.115
3	24	Female	Bachelor	60651	0.2	2606.510
4	64	Male	Master	81884	0.9	18984.780
...	...	...	...	...	...	...
995	42	Female	PhD	98170	0.7	17435.950
996	39	Male	High School	59174	0.9	11662.830
997	50	Male	Master	59808	0.4	7196.160
998	50	Female	Master	58272	0.7	13939.520
999	38	Female	Bachelor	32512	0.5	8312.800

1000 rows × 6 columns

```
In [ ]:
```

```
In [6]: dataset =pd.get_dummies(dataset,drop_first=True)
```

```
In [7]: dataset
```

```
Out[7]:
```

	age	income	purchase_frequency	spending	gender_Male	education_High School
0	42	53936	0.9	13227.120	False	True
1	49	82468	0.6	12674.040	False	False
2	55	56941	0.3	5354.115	True	False
3	24	60651	0.2	2606.510	False	False
4	64	81884	0.9	18984.780	True	False
...	...	...	...	...	...	...
995	42	98170	0.7	17435.950	False	False
996	39	59174	0.9	11662.830	True	True
997	50	59808	0.4	7196.160	True	False
998	50	58272	0.7	13939.520	False	False
999	38	32512	0.5	8312.800	False	False

1000 rows × 8 columns

```
In [8]: dataset.columns
```

```
Out[8]: Index(['age', 'income', 'purchase_frequency', 'spending', 'gender_Male',  
             'education_High School', 'education_Master', 'education_PhD'],  
            dtype='str')
```

```
In [9]: x = dataset[['age', 'income', 'purchase_frequency', 'gender_Male',  
                  'education_High School', 'education_Master', 'education_PhD']]
```

```
In [10]: x
```

```
Out[10]:
```

	age	income	purchase_frequency	gender_Male	education_High School	education_M
0	42	53936	0.9	False	True	
1	49	82468	0.6	False	False	
2	55	56941	0.3	True	False	
3	24	60651	0.2	False	False	
4	64	81884	0.9	True	False	
...	...	...	...	...	...	...
995	42	98170	0.7	False	False	
996	39	59174	0.9	True	True	
997	50	59808	0.4	True	False	
998	50	58272	0.7	False	False	
999	38	32512	0.5	False	False	

1000 rows × 7 columns

```
In [11]: y=dataset[['spending']]
```

```
In [12]: y
```

Out[12]:

	spending
0	13227.120
1	12674.040
2	5354.115
3	2606.510
4	18984.780
...	...
995	17435.950
996	11662.830
997	7196.160
998	13939.520
999	8312.800

1000 rows × 1 columns

```
In [13]: from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=11
)
```

```
In [14]: from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(x_train,y_train)
```

Out[14]:

▼ LinearRegression ⓘ ?

► Parameters

► Fitted attributes

```
In [15]: y_pred=regressor.predict(x_test)
y_pred
```

```
Out[15]: array([[ 4862.43499175],
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[-3489.93575583],
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[5280.93775212],
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[14344.81443198],
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```
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[11336.85515756],
[11115.78565937],
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[ 3234.44020863],
[13171.60561026],
[ 4715.93484045],
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[17750.65573593],
[13106.28758163],
[ -2184.24439321],
[16003.70167459],
[12104.28833361],
[ 3526.7424809 ],
[ 3268.52860134]])
```

In [16]: x_test

Out[16]:

	age	income	purchase_frequency	gender_Male	education_High School	education_M
25	56	36311	0.4	True	True	
464	28	71517	0.8	False	False	
372	50	82208	0.2	True	True	
730	45	27374	0.3	False	False	
757	21	32382	1.0	True	True	
...	...	...	...	...	...	...
278	22	36169	0.1	False	True	
449	50	52635	0.8	True	False	
221	47	41270	0.7	True	False	
741	51	38324	0.1	True	False	
572	48	32913	0.2	True	False	

200 rows × 7 columns

In [17]: y_test

Out[17]: **spending**

25	5526.220
464	13260.680
372	2822.080
730	6410.610
757	6619.100
...	...
278	880.845
449	16505.400
221	11944.450
741	2191.620
572	3329.130

200 rows × 1 columns

```
In [18]: from sklearn.metrics import r2_score
percentage=r2_score(y_test,y_pred)*100
print(f"result: {percentage} %")
```

result: 93.49355417351158 %

```
In [19]: import pickle

filename = "spending.sav"

pickle.dump(regressor, open(filename, "wb"))

loaded_model = pickle.load(open(filename, "rb"))
```

```
In [20]: age = int(input("Enter age: "))

gender = input("Enter gender (Male/Female): ").lower()

education = input("Enter education (High School/Bachelor/Master/PhD): ").

income = float(input("Enter income: "))

purchase_frequency = float(input("Enter purchase frequency: "))

if gender == "male":
    gender_male = 1
else:
    gender_male = 0

if education == "high school":
    education_high_school = 1
    education_master = 0
```

```

        education_phd = 0

    elif education == "master":
        education_high_school = 0
        education_master = 1
        education_phd = 0

    elif education == "phd":
        education_high_school = 0
        education_master = 0
        education_phd = 1

    elif education == "bachelor":
        education_high_school = 0
        education_master = 0
        education_phd = 0

    else:
        print("Invalid education")
        exit()

import pandas as pd
prediction_input = pd.DataFrame([[
    age,
    income,
    purchase_frequency,
    gender_male,
    education_high_school,
    education_master,
    education_phd
]], columns=[
    "age",
    "income",
    "purchase_frequency",
    "gender_Male",
    "education_High School",
    "education_Master",
    "education_PhD"
])

result = loaded_model.predict(prediction_input)

predicted_spending = result[0][0]

print(f"For {age} year old {gender.title()} with {education.title()} educ

```

For 52 year old Female with Phd education spending is 17709.40

In []: